

Counting the quadrant trees

A combinatorial derivation

$a(1) = 1$



$\cdot$

$\times 1$

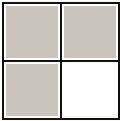
$a(2) = 6$

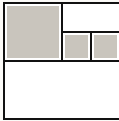


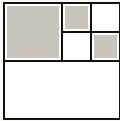
$\langle \bullet, \square, \square, \bullet \rangle$
 $\times 4$

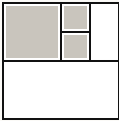
$\langle \bullet, \square, \bullet, \square \rangle$
 $\times 2$

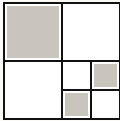
$a(3) = 76$

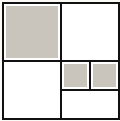
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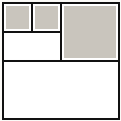
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 $\times 8$

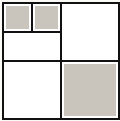
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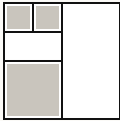
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 $\times 8$

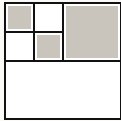
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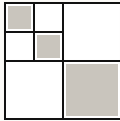
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 $\times 8$

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 $\times 4$

ORBIT AUDIT
 $4 + 8 + 8 + 8 + 8 + 4 + 8 + 8 + 8 + 8 + 8 + 4 = 76$

D_4 representatives; multipliers recover placements. Codes run CCW: NW $\rightarrow$ SW $\rightarrow$ SE $\rightarrow$ NE.

A quadtree square needs to be split into quadrants, possibly recursively, if it contains two or more points. Apply this procedure recursively to obtain a perfect quaternary tree pointing to n points on n true leaves. Forget those branches not ending at a true leaf point.

Now let i, j, k count nodes with $x = 2, 3$, or 4 valence, respectively, and distinguish branch-like (b_x) from true leaves (l_0). Each branch introduces a differential toward leaf count:

$$b_x : X \mapsto X' = X + x - 1$$

Summing over all branch nodes, starting from one root and ending in n true leaves we obtain that:

$$1 + i + 2j + 3k - n = 0$$

The total number of nodes including the root and true leaves will be $m = i + j + k + n$. The raw multinomial ordering of these symbols is:

$$\binom{m}{n, i, j, k} = \frac{m!}{n! i! j! k!}$$

Take any linear ordering of the b_x and l_0 symbols. Cyclically shift it until the partial sums from $X = 1$ reach zero only after the last term. This is the unique shift that reads as a valid rooted tree, so we divide by m .

The b_2 and b_3 branchings are associated to 6 and 4 distinct realizations in space, so we multiply by 6^i and 4^j to obtain the final closed-form summand:

$$\frac{(n+i+j+k-1)!}{n! i! j! k!} 6^i 4^j$$

The sum is taken over all non-negative indices subject to the zero-sum constraint:

$$a(n) = \sum \frac{(n+i+j+k-1)!}{n! i! j! k!} 6^i 4^j, \quad \text{where } i, j, k \geq 0, \quad 1 + i + 2j + 3k - n = 0.$$

The witness certificate

Just the integral-differential facts

INTEGRAL FORM

$$D(u)=1-6u-4u^2-u^3, \quad H_n(u)=\frac{1}{nu^nD(u)^n}, \quad q_n=\frac{1}{2\pi i}\oint H_n(u)du$$

Put B(u)=6u+4u²+u³, so D(u)=1-B(u). Expanding (1-B)⁻ⁿ and then B^m gives the Laurent series

$$q_n=\frac{1}{2\pi i}\oint \sum \sum \binom{n+m-1}{i,j,k}6^i4^ju^{i+2j+3k-n}du, \quad \text{where } m,i,j,k\geq 0, \ i+j+k=m.$$

A normalized contour integral around u=0 keeps exactly the u⁻¹ term:

$$\frac{1}{2\pi i}\oint u^rdu=1 \quad (r=-1), \quad 0 \quad (r\neq -1).$$

Thus the exponent condition moves underneath the sum:

$$i+2j+3k-n=-1 \iff 1+i+2j+3k-n=0,$$

$$q_n=\frac{1}{n}\sum \binom{n+m-1}{i,j,k}6^i4^j, \quad \text{where } m,i,j,k\geq 0, \ i+j+k=m, \ 1+i+2j+3k-n=0.$$

$$\frac{1}{n}\binom{n+m-1}{i,j,k}=\frac{(n+i+j+k-1)!}{n!i!j!k!} \implies q_n=a(n).$$

EXACT VERIFICATION IDENTITY

$$\sum_{r=0}^3P_r(n)H_{n+r}(u)=\frac{d}{du}(R(n,u)H_n(u))$$

RATIONAL CERTIFICATE

$$R(n,u)=\frac{N(n,u)}{u^2(u^3+4u^2+6u-1)^2}$$

$$\begin{aligned} N(n,u) &= n^2(64u^9+576u^8+2304u^7+4496u^6 \\ &\quad +2784u^5-5136u^4-8524u^3+204u^2+6396u-491) \\ &+n(112u^9+1008u^8+4032u^7+7988u^6 \\ &\quad +5784u^5-6228u^4-11872u^3-948u^2+6648u-491) \\ &+40u^9+360u^8+1440u^7+2960u^6 \\ &\quad +2960u^5+640u^4-440u^3+40u^2. \end{aligned}$$

ALGEBRAIC & DIFFERENTIAL OPERATORS (from OEIS A120593)

$$\begin{aligned} 5A(x) &= 4+x+A(x)^4 \iff Q=A-1, \quad Q=\frac{x}{D(Q)} \\ (256x^3+3072x^2+12288x-491)A^{(3)} &+ (1152x^2+9216x+18432)A^{(2)} \\ &+ (688x+2752)A^{(1)} - 40A = 0 \end{aligned}$$

REFERENCE

H. Brochet and B. Salvy, “Reduction-Based Creative Telescoping for Definite Summation of D-finite Functions,”

<https://arxiv.org/abs/2307.07216>

RECURRENCE OPERATOR

$$\sum_{r=0}^3P_r(n)q_{n+r}=0$$

q₀, ..., q₅ = 1, 1, 6, 76, 1201, 21252
n=0 is checked directly.

$$\begin{aligned} P_0 &= -8(4n+5)(2n+1)(4n-1) \\ P_1 &= -64(n+1)(48n^2+96n+43) \\ P_2 &= -6144(2n+3)(n+2)(n+1) \\ P_3 &= 491(n+3)(n+2)(n+1) \end{aligned}$$